

26/9/26

Perceptron

GrAD

"Assignment 1"

N:2

$$X = \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix}$$

$$W^1 = \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix}$$

$$W^2 = \begin{bmatrix} 0.44 & 0.03 & 0.55 \end{bmatrix}$$

$$W^3 = \begin{bmatrix} 0.88 \\ 0.71 \\ 0.29 \end{bmatrix}$$

$$W^4 = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix}$$

$$L.R = 0.5$$

$$O^1 = W^1 X = \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix} \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix}$$

$$O^1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix} \Rightarrow \text{Relu}(O^1) = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$O^2 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix} \begin{bmatrix} 0.44 & 0.03 & 0.55 \end{bmatrix}$$

$$\frac{dL}{dO^2} = \begin{bmatrix} 0.44 & 0.03 & 0.55 \end{bmatrix} \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$O^2 = [0.2972]$$

~~$$D^3 = \begin{bmatrix} 0.2972 & 0.55 \\ 0.71 & 0.29 \end{bmatrix} =$$~~

$$D^3 = \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix} \begin{bmatrix} 0.2972 \end{bmatrix} = \begin{bmatrix} 0.1634 \\ 0.2110 \\ 0.0861 \end{bmatrix}$$

$$D^4 = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.1634 \\ 0.2110 \\ 0.0861 \end{bmatrix}$$

$$D^4 = \begin{bmatrix} 0.3580 \\ 0.2826 \end{bmatrix}$$

Reconstruction Error:-

$$J(D) = - \sum_{d=1}^K x_d \log(x_d)$$

~~$$= -0.34 \log(0.5821) - 0.55 \log(0.369)$$~~

~~$$= 0.7419$$~~

$$= -0.34 \log(0.3580) - 0.55 \log(0.2826)$$

$$= 1.044$$

* Backward Pass:-

~~Forward Pass~~

$$\nabla O^1 \Delta v = O^1 - \text{actual}$$

$$= \begin{bmatrix} 0.3580 \\ 0.2826 \end{bmatrix} - \begin{bmatrix} 0.37 \\ 0.55 \end{bmatrix} = \begin{bmatrix} -0.018 \\ -0.2674 \end{bmatrix}$$

$$dw_1 = \begin{bmatrix} -0.018 \\ -0.2674 \end{bmatrix} \begin{bmatrix} 0.1634 & 0.2110 & 0.0861 \end{bmatrix}$$

$$= \begin{bmatrix} 0.002 & 0.0003 & 0.0001 \\ -0.043 & -0.056 & -0.023 \end{bmatrix}$$

$$\nabla O^3 \Delta v = \begin{bmatrix} 0.94 & 0.71 \\ 0.55 & 0.7 \\ 0.97 & 0.22 \end{bmatrix} \begin{bmatrix} -0.018 \\ -0.2674 \end{bmatrix}$$

$$= \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix}$$

$$dw_3 = \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix} \begin{bmatrix} 0.2972 \end{bmatrix} = \begin{bmatrix} -0.051 \\ -0.052 \\ -0.012 \end{bmatrix}$$

$$\nabla O^2 \Delta v = \begin{bmatrix} 0.55 & 0.71 & 0.29 \end{bmatrix} \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix}$$

$$= \begin{bmatrix} -0.234 \end{bmatrix}$$

$$\Delta w_2 = [-0.234] \begin{bmatrix} 0.5388 & 0.165 & 0.1005 \end{bmatrix} \\ = [-0.126 \quad -0.038 \quad -0.023]$$

$$\nabla O' \theta_{iv} = \begin{bmatrix} 0.44 \\ 0.03 \\ 0.05 \end{bmatrix} [-0.234] = \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix}$$

$$\Delta w_{i1} = \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix} \begin{bmatrix} 0.34 & 0.55 \end{bmatrix} \\ = \begin{bmatrix} -0.034 & -0.056 \\ -0.002 & -0.003 \\ -0.043 & -0.07 \end{bmatrix}$$

New Weights

$$w_4 = \begin{bmatrix} 0.969 & 0.548 & 0.969 \\ 0.731 & 0.728 & 0.231 \end{bmatrix}$$

$$w_3 = \begin{bmatrix} 0.575 \\ 0.736 \\ 0.296 \end{bmatrix}$$

$$w_{22} = \begin{bmatrix} 0.503 & 0.049 & 0.561 \end{bmatrix}$$

$$w_1 = \begin{bmatrix} 0.437 & 0.748 \\ 0.001 & 0.301 \\ 0.171 & 0.125 \end{bmatrix}$$

Random Pass

Q.3

$$Z_1 = y_1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$\mu' = [0.5 \ 0.5 \ 0.5] \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$\mu' = [0.402] = \mu$$

$$G' = [0.5 \ 0.5 \ 0.5] \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$G' = [0.402] = G$$

$$1) Z = \mu + G \cdot \epsilon \quad \because \epsilon = 1 \text{ (Assume)}$$

$$Z_2 = [0.402] + [0.402][1] = [0.804]$$

$$y_3 = Z_3 = \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix} [0.804] = \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$

$$Z_4 = \begin{bmatrix} 0.98 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$

$$y_4 = Z_4 = \begin{bmatrix} 0.968 \\ 0.764 \end{bmatrix}$$

$$L = -0.34 \ln(0.968) - 0.55 \ln(0.764)$$

$$L = 0.159$$

$$KL = -1/2 (1 + \log(6^2) - \mu^2 - \sigma^2)$$

$$KL = -1/2 (1 + \log(0.402^2) - 0.402^2 - 0.402^2)$$

$$KL = 0.572$$

$$J(Q) = L + KL = 0.159 + 0.572 = 0.731$$

In backward pass, $\nabla y^4 \partial v$, $\nabla z_4 \partial v$, dw_4 , $\nabla y^3 \partial v$, $\nabla z_3 \partial v$ will be calculated just like autoencoder. From dw_3 , all of them will be calculated as

$$dw_3 = \nabla z_3 \partial v \times z_2^T$$

$$\nabla z_2 \partial v = w_3^T \times \nabla z_3 \partial v$$

$$\nabla \mu \partial v = 1 \times \nabla z_2 \partial v + \mu$$

$$\nabla \mu' \partial v = \nabla \mu' \mu \times \nabla \mu \partial v \quad \because \nabla \mu' \mu = [1]$$

$$\nabla \sigma \partial v = \Sigma \times \nabla z_2 \partial v + (6 - 1/6)$$

$$\nabla \sigma' \partial v = [1] \times \nabla \sigma \partial v$$

$$dw_{\mu} = \nabla \mu' \Delta v \times y_1^T$$

$$dw_6 = \nabla b' \Delta v \times y_2^T$$

$$\nabla \mu' \Delta v = W_{\mu}^T \times \nabla \mu' \Delta v + W_6^T \times \nabla b' \Delta v$$

$$\nabla z_1 \Delta v = \nabla z_1 y' \times \nabla y_1 \Delta v$$

$$dw_1 = \nabla z_1 \Delta v \times X^T$$

$\nabla z_1 \Delta v$ and dw_1 will be calculated similarly as we done in autoencoder. Hence, we can update weights by using this method.